

INTEROPERABILITY MATURITY LAB (IML)

Technical Manuscript

Open Clinical Workspace, PostgreSQL Architecture, Clinical Reference Methodology and Traceable AI Evidence

| Document field                      | Current value                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------|
| Status                              | Open for Scientific and Technical Review                                      |
| Edition                             | Current consolidated technical edition - 6 September 2026                     |
| Supersedes                          | IML Technical Manuscript - 5 September 2026                                   |
| Companion document                  | IML Founding Manuscript - full consolidated working draft with AI-EVIDENCE-02 |
| Author                              | Dr Renato Barrios                                                             |
| Editorial and analytical assistance | OpenAI ChatGPT                                                                |
| Repository baseline                 | DRenatoBARRIOS/IML-contact, main branch, current public reference state       |

Status matters: implemented database structures, active workstreams, scaffolds and proposed future layers are distinguished throughout. No scaffold is presented as a clinically validated production service.

## Master Technical Index

| No. | Section                              | Technical scope                                                                                                |
|-----|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 01  | Scope, status and terminology        | Document boundaries, status vocabulary and relationship to the Founding Manuscript.                            |
| 02  | Engineering principles               | Health-first software, open source, vendor neutrality, replaceability, provenance and correction.              |
| 03  | Reference architecture               | Clinical core, terminology, integration layer, country packs, AI/evidence services and operations.             |
| 04  | PostgreSQL data foundation           | Schema separation, referential integrity, migrations, audit, backup, restoration and portability.              |
| 05  | Clinical data model                  | Identity, encounter, diagnostic assessment, observations/results, correction and provenance.                   |
| 06  | Terminology architecture             | Canonical terminology, KSH97-P, ICD-10, progressive mapping, translation review and ICD-11 readiness.          |
| 07  | Country adaptation                   | France and Guatemala packs, HPRIM-first France strategy and extension model.                                   |
| 08  | Interoperability layer               | Connector contracts, laboratory, imaging, FHIR/openEHR positioning, identity and national services.            |
| 09  | Deployment and operations            | macOS, Windows, Linux, local/remote PostgreSQL, offline capability, peripherals and recovery.                  |
| 10  | Audit, correction and Learning hooks | Traceability, versioning, correction propagation, documentary evidence and operational review.                 |
| 11  | Responsible clinical AI architecture | Human control, replaceable models, local extraction, de-identification and model-use traceability.             |
| 12  | AI-EVIDENCE-01                       | Implemented retrieval/provenance scaffold: schema, lifecycle, failure states, identifiers and source registry. |

| No. | Section                                                      | Technical scope                                                                                                                                                                 |
|-----|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | Evidence source contracts and clinical reference methodology | PubMed/NCBI, Europe PMC, ClinicalTrials.gov, Crossref, Cochrane public metadata, plus Harrison, Goldman-Cecil Medicine, CMDT, guideline triangulation and licensing boundaries. |
| 14  | AI-EVIDENCE-02                                               | Proposed appraisal, synthesis and clinical-translation layer with claim-to-source lineage.                                                                                      |
| 15  | Security, privacy and licensing                              | Secrets, minimisation, logging, access, rate limits, licences and failure containment.                                                                                          |
| 16  | Testing and validation                                       | Database, terminology, connectors, clinical safety, AI evaluation, resilience and regulatory review.                                                                            |
| 17  | Implementation status matrix                                 | What is active, scaffolded, proposed or awaiting validation.                                                                                                                    |
| 18  | Roadmap                                                      | Near-term implementation sequence and validation gates.                                                                                                                         |
| A-E | Appendices                                                   | Schema summaries, AI-EVIDENCE-02 object model, source registry, primary documentation and change log.                                                                           |

## 1 • Scope, Status and Technical Terminology

This manuscript is the engineering companion to the IML Founding Manuscript. It describes how the Open Clinical Workspace can be implemented as a modular, open-source and vendor-neutral clinical foundation while keeping national services, terminology, evidence retrieval and AI functions separable.

The document consolidates the previous 5 September 2026 Technical Manuscript and adds the current Open Clinical Workspace public architecture, the terminology canonicalisation work, the explicit AI service layer, the traceable AI-EVIDENCE-01 retrieval scaffold and the proposed AI-EVIDENCE-02 appraisal/synthesis layer.

The durable engineering objective is preservation of meaning. A user interface, hosting platform, external connector, AI model or terminology service may change; the clinical record, source provenance, correction history and scientific evidence trail should remain interpretable.

### 1.1 Status vocabulary

| Status   | Meaning in IML technical documentation                                                    |
|----------|-------------------------------------------------------------------------------------------|
| CONCEPT  | Architectural or scientific idea not yet represented as an implementable software object. |
| PROPOSED | Defined enough to review technically, but not yet implemented as a reference module.      |

| Status                             | Meaning in IML technical documentation                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| SCAFFOLD                           | Database structures, module registration or interface skeleton exist, but operational/clinical validation is incomplete.    |
| REFERENCE IMPLEMENTATION           | Runnable implementation work exists and can be tested; this does not itself establish clinical safety or regulatory status. |
| VALIDATED FOR RESEARCH USE         | Defined research workflow has passed specified validation; no automatic clinical-use claim.                                 |
| VALIDATED FOR DEFINED CLINICAL USE | Requires separate clinical safety, usability, governance and applicable regulatory validation for the stated intended use.  |

*These labels are intentionally conservative. A successful API call or a database migration does not convert a scaffold into a clinical service.*

## 1.2 Source hierarchy for this edition

- Current PostgreSQL migration 190\_ai\_evidence\_retrieval.sql for AI-EVIDENCE-01 is treated as the authoritative implementation source for the retrieval schema.
- Current migration 191\_terminology\_canonicalization.sql is treated as the authoritative implementation source for the terminology canonicalisation operation.
- The current public Clinical Workspace page is used to align the reference architecture, country-pack scope, deployment strategy and AI workstream wording.
- The full Founding Manuscript updated on 6 September 2026 is used as the scientific source for the proposed AI-EVIDENCE-02 architecture and for the status distinction between implementation and clinical validation.
- Where a component is part of the broader project design but not represented in the public repository snapshot, it is labelled workstream, proposed or scaffold rather than implemented.

*The clinical evidence methodology used by IML is a separate hierarchy from the software-source hierarchy above. It combines current guidelines, licensed clinical reference texts, systematic reviews and recent primary literature rather than treating bibliographic search as a complete clinical reasoning method.*

## 2 • Engineering Principles

### 2.1 Health-first software

The workspace is not an end in itself. Every module should justify its presence through a clinical, public-health, scientific or continuity purpose. Feature count is not a maturity metric. A small, inspectable and reliable installation may be preferable to a complex platform that cannot be maintained or corrected.

### 2.2 Open source without naïve assumptions

Open-source code improves inspectability and autonomy but does not automatically provide clinical quality, security, maintenance or governance. IML therefore treats open source as an enabling property that must be combined with reproducible installation, lifecycle support, testing, documentation, correction and accountable stewardship.

### 2.3 Vendor neutrality and replaceable components

Clinical data, terminology, integration interfaces, evidence metadata and AI services should be separated behind explicit contracts. The architecture should permit replacement of a model provider, terminology server, cloud service or connector without silently rewriting clinical meaning.

## 2.4 Provenance before automation

Every important transformation should be attributable. For imported results this includes source system and payload provenance. For terminology it includes source release and mapping provenance. For AI it includes the function, model or service, input/output event, evidence set and human validation state.

## 2.5 Correction as a first-class operation

Clinically meaningful data should not be silently overwritten. Corrections should preserve history, responsible actor, timestamps and the relationship between old and new representations. This is both a software-quality requirement and a direct implementation of the IML Learning domain.

## 2.6 Progressive deployment

The workspace should provide a useful first installation without requiring a complete national architecture. Local database operation, bounded connectors and offline-capable workflows permit incremental adoption in constrained settings while keeping a path toward broader interoperability.

# 3 • Reference Architecture

| Layer                    | Primary technical responsibilities                                                                               | Isolation objective                                                                               |
|--------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Clinical core            | Patient context, encounter, clinical history, diagnostic assessment, observations/results, audit and correction. | Must remain usable without a particular national service or AI provider.                          |
| Identity and trust       | Patient/professional identifiers, authentication context, provenance and access decisions.                       | Use existing identity systems; do not create a new universal identity number.                     |
| Terminology services     | Canonical concepts, designations, releases, mappings, provenance, review and search.                             | Keep source terms and mapped concepts distinguishable and versioned.                              |
| Open integration layer   | Laboratory, imaging, registry, national identity, consent, messaging and other external connectors.              | Vendor/national assumptions remain outside the common clinical model.                             |
| Country packs            | Language, national terminology, billing/reimbursement, messaging, identity and regulatory adapters.              | Adding a country should not require forking the common core.                                      |
| AI and evidence services | Clinical evidence retrieval, local extraction, terminology assistance and future synthesis.                      | AI remains assistive and replaceable; durable data and evidence provenance survive model changes. |
| Operations               | Deployment, backups, restoration, offline/synchronisation, observability and peripherals.                        | Care should not collapse when the ideal network or cloud service is unavailable.                  |

## 3.1 Architectural boundary rule

A national connector may know about a French identifier or reimbursement service. The clinical core should know only the normalized patient, encounter or administrative relationship required by the common model. Similarly, an AI service may receive an authorised, proportionate context, but the clinical database must not be defined by a single model provider.

## 3.2 Data flow principle

1. Capture or receive information in its source context.
2. Preserve the original source/provenance needed for audit and reprocessing.
3. Normalize into the common model using explicit terminology and validation rules.
4. Expose the normalized record to authorised clinical workflows.
5. Allow correction without deleting the historical source trail.
6. Permit derived analytics or AI assistance through a separate, auditable service boundary.

## 3.3 Reference data flow for evidence-aware clinical assistance

Clinical context -> de-identified evidence question -> source-by-source evidence retrieval -> canonical evidence set -> human-reviewed appraisal/synthesis -> clinical or research-facing presentation. AI-EVIDENCE-01 covers the retrieval/provenance segment. AI-EVIDENCE-02 is proposed for appraisal, synthesis and translation.

# 4 • PostgreSQL Reference Data Foundation

## 4.1 Why PostgreSQL

PostgreSQL is the current reference relational foundation because it offers transactions, referential integrity, constraints, structured JSON metadata, indexing, cryptographic extensions, mature backup/restore tooling and an inspectable SQL migration model. The choice is pragmatic rather than ideological.

## 4.2 Schema separation

| Schema / namespace      | Technical role evidenced or referenced in current work                             |
|-------------------------|------------------------------------------------------------------------------------|
| iml_clinical            | Clinical encounter and diagnostic-assessment context referenced by AI-EVIDENCE-01. |
| iml_identity            | Practitioner identity referenced for attributable evidence-query requests.         |
| iml_terminology         | Canonical active terminology environment after migration 191.                      |
| iml_terminology_staging | Controlled staging area for terminology imports/mappings.                          |
| iml_ai                  | AI evidence retrieval and provenance structures created by migration 190.          |
| iml_system              | Module registry used to declare technical module status/version.                   |

*Schema separation improves comprehensibility but is not a security boundary by itself. Database privileges, roles and application-layer authorisation remain necessary.*

## 4.3 Migration discipline

Migrations should be additive whenever possible, idempotent where practical and equipped with assertions that fail safely. Migration 191 provides an explicit example: it refuses to retire a legacy terminology schema if it contains data, and it refuses to rename staging structures when the business content differs materially from the canonical staging content.

## 4.4 Canonicalisation without destructive cleanup

The terminology migration deliberately renames legacy schemas instead of dropping them. This preserves auditability and reduces the risk that a cleanup operation destroys information that later proves important. Similar conservative rules should be applied to future clinical and AI migrations.

## 4.5 Integrity and foreign keys

AI-EVIDENCE-01 links evidence queries to clinical encounters or diagnostic assessments using nullable foreign keys and links the requester to a practitioner identity. Source runs, identifiers and query-result relations are similarly constrained so that provenance is represented structurally rather than left to free text.

## 4.6 Backup, restoration and reversibility

Reference deployment testing should include full backup, restoration into a clean environment, version-compatible migration, export of clinically meaningful data, recovery after partial failure and verification that identifiers and provenance remain intact after restore.

## 4.7 Hashing and fingerprints

AI-EVIDENCE-01 uses SHA-256 fields for the question/search expression and optional request/response fingerprints. Fingerprints support reproducibility and duplicate/failure analysis without requiring storage of unnecessary raw sensitive content. Hashes are provenance aids, not substitutes for secure access control.

# 5 • Clinical Data Model

## 5.1 Patient and professional identity

The workspace should support local and national patient identifiers, duplicate prevention, provenance and correction pathways. Professional identity should support attributable clinical and technical actions. The architecture should integrate recognised identity systems rather than invent a new universal person number.

## 5.2 Encounter as the clinical unit of work

An encounter anchors patient context, consultation reasons, observations, assessments and evidence questions. It allows a later reviewer to distinguish information recorded in one episode from information imported or inferred later.

## 5.3 Diagnostic assessment

Diagnostic assessment should remain distinct from raw observations. A culture result, a clinician interpretation and a treatment decision are different objects. AI-EVIDENCE-01 can associate a retrieval question with a diagnostic assessment without conflating the retrieved literature with the diagnosis itself.

## 5.4 Results and laboratory provenance

Structured results should preserve source laboratory, acquisition/collection time, original identifiers or payload reference, units, reference ranges where applicable, terminology mapping and correction state. A normalized result must remain traceable to the source representation.

## 5.5 Clinical history and correction

Longitudinal information should remain attributable, versioned and correctable. A later correction should not erase the fact that a prior version existed and may have influenced earlier decisions.

## 5.6 AMR/BMR test case

The MDR E. coli UTI demonstrator is a useful stress test of the clinical model because laboratory detection, symptomatic infection, colonisation, asymptomatic bacteriuria and contamination must remain distinguishable. The minimum model must therefore carry symptoms, specimen context, microbiology, treatment and outcome rather than only an organism code.

## 6 • Terminology Architecture

### 6.1 Canonical terminology environment

Migration 191 establishes `iml_terminology` and `iml_terminology_staging` as the canonical active terminology namespaces and conservatively archives older schema names. Canonicalisation is designed to prevent parallel active terminology stores from drifting silently.

### 6.2 KSH97-P for concise consultation reasons

KSH97-P remains the initial primary-care terminology workstream for concise reasons for consultation. The intended workflow allows a small number of focused consultation reasons while more extensive or formal classification can use ICD-10. The objective is to preserve clinically usable language without hiding mappings.

### 6.3 ICD-10 and national variants

ICD-10 and national variants support classification, reporting and country requirements. Each imported release should retain authority, version and mapping provenance. Mappings should represent ambiguity and one-to-many relationships where necessary rather than forcing false equivalence.

### 6.4 Mapping model

1. Preserve the source terminology and source expression.
2. Record the target terminology release and candidate mapping expression.
3. Allow analysis of exact, broader, narrower, composite or unresolved relationships.
4. Require review status for clinically consequential mappings.
5. Keep previous mapping versions available for audit and historical record interpretation.

### 6.5 Translation workflow

Translation should use a review queue rather than auto-publishing model output as an official designation. French, Latin American Spanish and later Guatemala-specific terms can be proposed by software or AI, but promotion to a canonical designation requires explicit validation and source authority metadata.

### 6.6 ICD-11 readiness

ICD-11 readiness is treated as a future multi-classification/shadow-coding capability. It should not be presented as live ICD-11 clinical coding until release acquisition, mapping strategy, terminology services, user-interface behaviour and local regulatory/operational validation are complete.

### 6.7 AI assistance in terminology

AI can accelerate terminology search, candidate mapping generation or translation review. It must not silently mutate canonical terminology. Candidate suggestions should retain model/tool provenance and a deterministic human-validation outcome.

## 7 • Country Adaptation Strategy

### 7.1 Common core, country-specific edges

Country adaptation belongs around, not inside, the common clinical model. Language, national identifiers, reimbursement, authentication, terminology releases and messaging services vary. A country pack should declare these dependencies explicitly so that a new country is additive rather than a fork.



## 7.2 France pack

The current France scope includes INS and patient identity, MSSanté, DMP/Mon espace santé, CPS/e-CPS, Sesam-Vitale, SCOR, Ségur-related constraints, prescribing/LAP requirements, NGAP, CCAM, CIM-10-FR, LOINC FR, ALD, C2S, AME, AT/MP, laboratory and imaging integration.

## 7.3 HPRIM-first laboratory pathway

HPRIM laboratory results are the first controlled France-specific integration target. The technical goal is to preserve original message/source provenance, resolve patient/encounter context, normalize analytes/units where safe, map terminology explicitly, detect duplicates and support correction.

## 7.4 Guatemala pack

Guatemala is the second reference adaptation. The first software objective is a Spanish-language workspace while preserving the same common model. Intermittent connectivity, local deployment and maintainability are first-class design constraints rather than afterthoughts.

## 7.5 Country-pack contract

| Country-pack concern | Required declaration                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------|
| Language             | UI languages, clinical designation languages and translation-validation process.              |
| Identity             | National/local patient and professional identity mechanisms and mapping rules.                |
| Terminology          | Applicable national releases, local codes and mapping authority.                              |
| Messaging            | Laboratory, referral, discharge, secure messaging and national service connectors.            |
| Financing            | Billing, coding, entitlement or payer interoperability rules where applicable.                |
| Regulation           | Clinical-software, data-protection, AI and security obligations relevant to intended use.     |
| Deployment           | Connectivity assumptions, server/workstation constraints, offline strategy and support model. |

# 8 • Interoperability Layer

## 8.1 Connector principle

A connector translates an external contract into the common IML model and preserves enough source material/provenance for troubleshooting and reprocessing. Vendor-specific fields should not leak into unrelated clinical modules unless they represent clinically necessary information.

## 8.2 FHIR positioning

FHIR is a strong candidate for resource-oriented APIs and exchange contracts. IML should use it where it improves interoperability and ecosystem compatibility, while avoiding the assumption that every source must be transformed into FHIR before a useful first integration can exist.

## 8.3 openEHR positioning

openEHR provides useful ideas for durable clinical modelling and archetype-governed structures. IML can use these ideas or interfaces selectively while maintaining its own explicit PostgreSQL reference model.

## 8.4 DICOM and imaging

Imaging should preserve DICOM-compatible transport and identifiers where relevant. The clinical report and interpretation remain separate from image transport. Open components such as Orthanc can be considered as reference tools where they fit local architecture.

## 8.5 Identity, consent and national services

Connectors for identity, consent/lawful basis and national services should be explicit service boundaries. A failure in a national service should not corrupt the local clinical record, and local operation should make the dependency visible to the user.

## 8.6 Import-on-demand and minimum necessary data

The workspace should retrieve information for a defined purpose rather than replicate everything by default. Import-on-demand, controlled caching and clear retention rules reduce unnecessary data exposure and support offline operation when needed.

## 8.7 Connector observability

Every connector should expose success, partial success, validation failure, timeout, remote unavailability and local rejection as distinct states. Observability is required so that clinical users and technical staff can distinguish missing information from a failed exchange.

# 9 • Deployment and Operations

| Deployment concern        | Reference position                                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| macOS                     | Development and professional-workstation target where appropriate.                                                       |
| Windows                   | Required target because many health workstations and local peripherals depend on Windows.                                |
| Linux                     | Reference server environment; Debian as conservative baseline and Ubuntu LTS as secondary target.                        |
| Local PostgreSQL          | Supports site autonomy and weak-connectivity environments.                                                               |
| Remote PostgreSQL/API     | Supports managed and multi-site deployments.                                                                             |
| Offline-capable workflows | Clinical operation should continue during intermittent connectivity; synchronization/conflict handling must be designed. |
| Peripherals               | Health-card readers and other local devices are anticipated rather than excluded by cloud-only assumptions.              |

## 9.1 Local versus remote data

A deployment may keep data locally, use a remote database, or combine local operation with controlled synchronization and APIs. The choice should be documented as an operational and governance decision, not hidden inside application defaults.

## 9.2 Offline capability

Offline capability requires more than a local cache. The system must define which functions remain available, how writes are queued, how identity is verified, how conflicts are resolved and how users are informed when information may be stale.

## 9.3 Backup and recovery

Backup schedules, retention, encryption, restoration procedures and recovery objectives should be testable. A backup that has never been restored is not demonstrated resilience.

## 9.4 Observability

Technical logs should cover application health, migrations, connector failures, queue depth, synchronization state and AI evidence-source failures without copying unnecessary patient-identifying data into general logs.

## 9.5 Update and dependency discipline

The open-source toolchain should pin or document versions, track security advisories and make dependency updates reproducible. Unsupported operating systems, libraries or database versions should be treated as clinical continuity risks when they support active care.

# 10 • Audit, Correction and Learning Hooks

## 10.1 Attributable actions

Meaningful clinical and administrative actions should identify the responsible practitioner, system or process. Automated events should be distinguishable from human edits. Model-generated candidates should be distinguishable from signed clinical facts.

## 10.2 Versioned correction

Correction should produce a new state linked to the previous state rather than erase history. Systems reusing corrected information should be able to detect and propagate the correction where appropriate.

## 10.3 Source correction versus interpretation correction

An imported laboratory value may be corrected by the source laboratory; a clinician may separately revise the interpretation. These events should not be collapsed. Provenance must show which layer changed.

## 10.4 Learning-domain technical hooks

| Learning criterion                   | Technical implication                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| LRN-1 Error detection/reporting      | Issue/event capture, validation errors, incident identifiers and review workflow.                                    |
| LRN-2 Continuous improvement         | Metrics, incident trend analysis, post-change evaluation and reproducible configuration/version records.             |
| LRN-3 Traceable correction and reuse | Versioned data, correction lineage and propagation to downstream reuse.                                              |
| LRN-4 Documentary reliability        | Versioned public/technical documentation, stable references, dates and correction history.                           |
| LRN-5 Institutional answerability    | Evidence register capable of recording authoritative findings, scope, response/follow-through state and limitations. |

## 10.5 Audit minimisation

Auditability does not justify indiscriminate logging. Logs should record enough to reconstruct consequential events without duplicating full clinical narratives or secrets into operational systems.

# 11 • Responsible Clinical AI Architecture

## 11.1 AI is a service layer, not the source of truth

The patient record remains the clinical source of truth. AI can retrieve, structure, suggest, compare or explain. It must not silently rewrite diagnoses, results or signed decisions.

## 11.2 Replaceable models

Local models, university-hosted services and controlled external providers should be replaceable behind explicit interfaces. Durable clinical data and scientific provenance must survive provider changes.

## 11.3 Human clinical control

AI outputs should have a validation state. Candidate facts, terminology suggestions, evidence summaries and decision-support outputs should remain reviewable and distinguishable from validated data.

## 11.4 Local structured clinical extraction

Local extraction can identify candidate facts from dictation or observation for clinician review. The conservative design principle is candidate generation rather than silent chart insertion. Local processing may reduce external disclosure but still requires model validation, audit and correction.

## 11.5 Traceable model use

For consequential functions, the system should record enough context to identify the AI function, model/service version, timestamp, relevant input snapshot or fingerprint, output fingerprint, human reviewer and final action. The clinical record should not become an unreadable model log, so detail may be stored in a linked provenance object.

## 11.6 De-identification

External evidence retrieval should use de-identified search expressions. Direct patient identifiers and unnecessary narrative should not be sent to bibliographic services. De-identification reduces risk but does not eliminate all re-identification or linkage risk.

## 11.7 Regulatory boundary

The intended use determines regulatory consequences. A research evidence-search tool, a clinician-facing literature summary and a patient-specific autonomous recommendation are not the same product claim. IML should define and validate each intended use separately before clinical deployment.

# 12 • AI-EVIDENCE-01 • Detailed Retrieval and Provenance Architecture

AI-EVIDENCE-01 v0.1.1 is the implemented PostgreSQL scaffold for traceable bibliographic evidence retrieval. It is additive to the existing IML core and creates the `iml_ai` schema. The current module registry identifies the module as `AI_EVIDENCE_RETRIEVAL` with status `SCAFFOLD` and version `AI-EVIDENCE-01/0.1.1`.

## 12.1 Core objects

| Object                         | Key role                                       | Important implementation properties                                                                                                                   |
|--------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| iml_ai.evidence_source         | Registry of available evidence sources.        | source_code primary key; source_kind/access_mode checks; URLs, enabled flag, full-text policy, capabilities, licence note; credentials excluded.      |
| iml_ai.evidence_query          | One evidence question/search request.          | Optional encounter_id and assessment_id links; question/search hashes; question type; language; filters; lifecycle status; requested_by practitioner. |
| iml_ai.evidence_source_run     | One execution of one source for one query.     | Separate run status; request/response fingerprints; result count; error code/summary; timing; metadata.                                               |
| iml_ai.evidence_record         | Canonical bibliographic/scientific record.     | Unique canonical_key; record type; publication metadata; OA/retraction flags; metadata JSON.                                                          |
| iml_ai.evidence_identifier     | Persistent/source identifiers.                 | PMID, PMCID, DOI, NCT, COCHRANE or OTHER; source URL; retrieval time; raw metadata hash.                                                              |
| iml_ai.evidence_query_result   | Relation between query, source run and record. | Source rank, relevance score, retrieval/inclusion state, exclusion reason, retrieval metadata.                                                        |
| iml_ai.evidence_access_profile | Deployment-level access capability.            | Public/institutional/link-resolver/disabled modes; configuration without passwords, cookies or secrets.                                               |

## 12.2 Question types

The current schema supports diagnostic, treatment, prognosis, harm, screening, monitoring and other evidence-question types. This is deliberately broader than treatment alone and supports later structured retrieval strategies.

## 12.3 Query lifecycle

1. A clinical workflow identifies an evidence question associated, where appropriate, with an encounter or diagnostic assessment.
2. The application derives a de-identified search expression and stores its SHA-256 fingerprint.
3. One source run is created for every evidence source actually invoked.
4. Each source run records its own state, timing, counts, fingerprints and error information.
5. Retrieved records are normalized into canonical evidence\_record objects and linked to source identifiers.
6. Query-result relations preserve source-specific rank, inclusion/exclusion state and retrieval time.
7. Later summarisation or appraisal consumes the durable record set rather than replacing it.

## 12.4 Failure states are first-class data

The current run-status model distinguishes queued, running, success, no\_results, partial, unavailable, rate\_limited, failed and cancelled. This is essential because no evidence found is not the same as evidence source failure. The UI and any later synthesis layer must preserve that distinction.

## 12.5 Canonical records and deduplication

Canonical records allow the same paper or trial to be referenced from more than one source without creating multiple independent truths. Strong identifiers should be preferred for deduplication. Title-only matching should be a cautious fallback because titles can change or collide.

## 12.6 Full-text boundary

AI-EVIDENCE-01 does not store publisher full text by default. The initial layer is designed around metadata, identifiers, abstracts where permitted and lawful access links. Full-text retrieval should be added only with explicit licensing, access-control and provenance rules.

## 12.7 Retraction and update handling

The evidence\_record object includes a retracted flag and structured metadata. Future connector workflows should check corrections, retractions or material metadata updates and preserve when the change was learned, rather than silently mutating historical synthesis context.

## 12.8 Access profiles

Institutional access is represented as a capability profile, not as stored credentials. Passwords, session cookies, API keys and institutional secrets belong in deployment-specific secret management outside clinical evidence tables.

# 13 • Evidence Source Contracts

## 13.1 PubMed / NCBI

Role: primary biomedical bibliographic discovery and identifier resolution through NCBI Entrez E-utilities. The scaffold records PubMed/NCBI as a bibliographic database accessed through a public API, with PMID, PMCID and DOI capabilities.

- Reference API base in the current scaffold: <https://eutils.ncbi.nlm.nih.gov/entrez/eutils/>.
- Preserve the exact de-identified search expression, retrieval timestamp and source execution state.
- Use PMID as a strong source identifier and resolve PMCID/DOI when available.
- Respect NCBI usage guidance, identify the client appropriately and implement backoff/caching rather than aggressive repeated queries.
- Presence in PubMed is bibliographic evidence of indexing, not a quality rating or clinical recommendation.

## 13.2 Europe PMC

Role: complementary biomedical discovery, metadata and open-access link discovery. Europe PMC overlaps heavily with PubMed but is not identical and can add PMCID/open-access relationships and broader metadata.

- Reference API base: <https://www.ebi.ac.uk/europepmc/webservices/rest/>.
- Deduplicate against PubMed using strong identifiers before title heuristics.
- Preserve Europe PMC as a distinct source run even when the canonical record is shared.
- Open-access links should preserve licence/access metadata where available rather than imply universal reuse rights.

## 13.3 ClinicalTrials.gov

Role: trial registration and results-status metadata. Registry evidence answers different questions from a published article and should remain typed as trial registration/protocol/results status rather than silently merged with peer-reviewed findings.

- Reference API base: <https://clinicaltrials.gov/api/v2/>.
- Use NCT identifiers as strong registry identifiers.
- Preserve trial status, dates and available results metadata separately from article publication records.

- Compare registry and publication evidence where useful, but do not treat registration as proof of outcome.

## 13.4 Crossref

Role: DOI-centred scholarly metadata enrichment and persistent identifier resolution. Crossref is useful for publisher/journal/funder/licence/update metadata and deduplication, not as an independent clinical quality source.

- Reference API base: <https://api.crossref.org/>.
- Use DOI as the primary persistent scholarly identifier when available.
- Respect Crossref REST etiquette and cache stable metadata responsibly.
- Preserve source-specific retrieval timestamps because metadata can be corrected over time.

## 13.5 Cochrane public metadata

Role: systematic-review and protocol discovery through public metadata and lawful link-out. The current scaffold explicitly labels the source as public metadata rather than unrestricted full text.

- Retain Cochrane/DOI identifiers and source URLs where available.
- Subscription or institutional full-text access is a separate capability from public metadata discovery.
- Do not copy restricted review text into local stores without appropriate rights.
- Do not describe an IML synthesis as a Cochrane review unless it actually follows Cochrane methods and governance.

## 13.6 Source orchestration

The sources should not be treated as interchangeable search engines. PubMed and Europe PMC support literature discovery; ClinicalTrials.gov provides registry status; Crossref enriches DOI metadata; Cochrane supports systematic-review discovery. The orchestrator should record why a source was queried and preserve source-specific failure states.

## 13.7 Licensed Clinical Reference Methodology: Harrison, Cecil and CMDT

IML does not treat PubMed or any bibliographic database as a substitute for clinical medicine. The evidence workflow begins with a clinically framed question and deliberately combines current recommendations with established clinical reference syntheses and the recent research literature.

The licensed clinical reference layer includes Harrison's Principles of Internal Medicine, Goldman-Cecil Medicine and Current Medical Diagnosis & Treatment (CMDT). In earlier working discussions, the latter was sometimes abbreviated CMCT; the manuscript uses the standard abbreviation CMDT. These sources are used because they organise disease definitions, pathophysiology, differential diagnosis, clinical presentation, diagnostic strategy, red flags and management in a form designed for clinical reasoning rather than bibliographic retrieval alone.

The three references are complementary rather than interchangeable. Harrison provides a broad internal-medicine synthesis and pathophysiological framework. Cecil provides an independent comprehensive clinical synthesis useful for comparison and for exposing differences in emphasis. CMDT provides a deliberately practical, frequently updated diagnostic and therapeutic orientation. None is treated as infallible or as an automatic replacement for current specialty guidelines.

## 13.8 IML Clinical Evidence Sequence

The reference workflow is intentionally ordered. It is designed to avoid two opposite errors: relying only on a textbook that may lag behind a new recommendation, or relying only on the newest paper without the clinical synthesis needed to interpret it.

1. Clinical question and patient context. Define the actual clinical problem, relevant population, uncertainty and intended decision before searching.
2. Current clinical guidelines. Identify the most relevant current recommendations from competent professional societies, public authorities or recognised guideline bodies, recording version and date.
3. Licensed clinical reference synthesis. Review Harrison, Cecil and CMDT for disease framing, differential diagnosis, diagnostic thresholds, red flags, treatment structure and points that a narrow literature search may miss.
4. Systematic reviews and meta-analyses. Use Cochrane and other high-quality systematic evidence to test the broader evidence base and identify consistency, heterogeneity and uncertainty.

5. Recent primary literature. Search PubMed, Europe PMC, trial registries and related sources for important evidence published after the reference texts or guideline evidence cutoff, including trials, safety signals and major observational studies.
6. Comparative synthesis. Compare recommendations, reference-text synthesis and recent evidence explicitly. Preserve agreement, disagreement, recency and applicability rather than averaging them into one opaque answer.
7. Clinician review. A qualified professional reviews the evidence trail and determines whether the synthesis is appropriate to the patient, setting and intended decision.

### 13.9 Triangulation and Disagreement Rules

Concordance is informative but not proof. If a current guideline, Harrison, Cecil, CMDT and recent systematic evidence agree, IML may describe the position as convergent while still preserving the individual sources.

Disagreement is a first-class result. If Harrison, Cecil and CMDT differ, the system should show the divergence, edition dates and relevant chapter context. The AI must not silently choose one text or manufacture a consensus.

When a current specialty guideline conflicts with an older textbook on a management recommendation, the newer guideline normally receives greater operational weight if it is applicable to the patient and supported by an appropriate evidence process. The older reference remains useful for pathophysiology, differential diagnosis and historical context.

When recent primary literature appears to challenge a current guideline or textbook, it should be labelled emerging evidence unless the body of evidence and methodological quality justify stronger wording. Novelty alone does not outrank synthesis.

Where guidelines disagree with one another, IML should record the organisations, dates, populations, methods and recommendation strength rather than flattening the conflict.

For all source classes, applicability matters. A recommendation for a trial population, healthcare setting or national formulary should not be transferred automatically to a different patient or country.

### 13.10 Clinical Knowledge Matrix

For a defined clinical question, IML should be able to construct a source-by-source matrix. The matrix is not a score of which source is best; it is a transparent comparison of what each source contributes.

| Source class                      | Primary contribution                                       | Fields to compare                                                                                 | Typical caution                                                        |
|-----------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Current guideline                 | Current recommendation and decision thresholds             | Recommendation, strength, population, date, evidence basis                                        | May be narrow, context-specific or methodologically heterogeneous      |
| Harrison                          | Internal-medicine synthesis and pathophysiological framing | Definition, mechanisms, differential diagnosis, diagnostic strategy, management overview          | Edition cycle may lag very recent practice changes                     |
| Goldman-Cecil Medicine            | Independent comprehensive clinical synthesis               | Presentation, differential diagnosis, work-up, treatment framing, complications                   | Edition cycle and editorial emphasis may differ from guidelines        |
| CMDT                              | Practical current diagnostic and treatment orientation     | Diagnostic approach, treatment options, dosing/monitoring context where licensed text supports it | Must still be checked against specialty guidance and local formulary   |
| Systematic review / meta-analysis | Structured synthesis of a defined evidence body            | Population, intervention, outcomes, heterogeneity, certainty/limitations                          | Can be outdated or include heterogeneous studies                       |
| Recent primary literature         | Newest potentially practice-changing evidence              | Design, population, effect, harms, date, applicability                                            | Single studies should not automatically override established synthesis |



| Source class   | Primary contribution              | Fields to compare                                               | Typical caution                                          |
|----------------|-----------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| Trial registry | Ongoing/unpublished study context | Status, prespecified outcomes, completion, results availability | Registration is not equivalent to peer-reviewed evidence |

### 13.11 Licensing and Access Boundary for Harrison, Cecil and CMDT

Harrison, Cecil and CMDT are subscription or licensed clinical references. Their importance to the methodology does not grant IML a right to scrape, copy or store their full text.

The default technical model is therefore licensed human access and citation/link-out. The workspace may record bibliographic metadata, edition, chapter reference, access provenance and a clinician-authored note or structured conclusion, but it should not ingest copyrighted full text unless a publisher or institutional licence explicitly permits machine retrieval, processing and storage.

Institutional credentials, cookies, passwords and access tokens must remain outside the evidence database. Machine access to a licensed reference should be enabled only under a documented contract or API entitlement that permits the intended operation.

Where machine rights are absent, AI may still help the clinician compare manually supplied, appropriately limited notes or citations with open bibliographic evidence, but it must not bypass access controls or reconstruct copyrighted chapters.

The provenance record should distinguish open bibliographic metadata, open-access full text, institutionally licensed human reference and contractually authorised machine-readable reference. This distinction is required for reproducibility and licensing compliance.

### 13.12 Integration with AI-EVIDENCE-01 and AI-EVIDENCE-02

AI-EVIDENCE-01 remains the traceable retrieval layer for open or appropriately accessible evidence sources. Its current registry covers PubMed/NCBI, Europe PMC, ClinicalTrials.gov, Crossref and Cochrane public metadata. The clinical reference methodology extends the source model conceptually to licensed-reference metadata and clinician-verified extracts without pretending that subscription textbooks are ordinary public APIs.

AI-EVIDENCE-02 should consume not only a list of papers but a typed evidence set. A current guideline, a Harrison chapter, a Cecil chapter, a CMDT section, a Cochrane review and a new randomised trial have different epistemic roles. The synthesis layer should preserve those roles instead of converting every source into an undifferentiated vector of text.

The proposed claim graph should therefore permit each clinical claim to identify whether it is supported, contradicted or qualified by guidelines, clinical reference syntheses, systematic reviews or primary studies. Source date, edition, applicability and reviewer status should remain visible.

The resulting output is not a generic literature summary. It is a clinically organised, provenance-preserving synthesis designed to support professional reasoning, differential diagnosis, treatment review and explicit identification of uncertainty.

*Methodological invariant: IML must never replace the clinical synthesis layer with a PubMed-only workflow. Bibliographic retrieval supplies evidence; clinical references and guidelines organise that evidence into medicine; clinician review remains responsible for the final interpretation.*

## 14 • AI-EVIDENCE-02 • Proposed Appraisal, Synthesis and Clinical Translation Architecture

*AI-EVIDENCE-02 is proposed as the next layer after retrieval. It is not implemented or clinically validated in the current repository baseline. Its purpose is to define the provenance, audit and human-validation requirements before any synthesis engine is allowed to influence scientific or clinical decisions.*

### 14.1 Why it is a separate layer

Retrieval answers what evidence was found, from which source, under which query and with which technical outcome. Appraisal introduces qualitatively different judgements: study design, relevance, risk of bias, indirectness, missingness,

heterogeneity and certainty. These decisions require methods and human expertise that should not be hidden inside the retrieval layer.

### 14.1.1 Typed clinical evidence stack

AI-EVIDENCE-02 should treat guideline recommendations, licensed clinical reference synthesis, systematic reviews and primary studies as distinct source classes with different roles. The synthesis engine should therefore preserve source type, publication or edition date, applicability, licensing status and reviewer validation. Harrison, Cecil and CMDT belong to the clinical-synthesis layer, not to the public bibliographic retrieval layer.

## 14.2 Claim graph rather than one answer

A synthesis should be represented as a set of atomic claims connected to supporting, contradicting or non-informative records. This makes disagreement visible, allows later updates and prevents the user interface from turning a complex evidence base into one opaque generated paragraph.

## 14.3 Proposed object model

| Proposed object             | Purpose                                                                                               | Minimum provenance expectation                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| evidence_review_question    | Versioned review/clinical question, PICO-like framing where useful, outcomes, scope and intended use. | Author/reviewer, version, creation time, question fingerprint.                         |
| evidence_eligibility_rule   | Inclusion/exclusion criteria used for screening.                                                      | Versioned rules; change history; responsible reviewer.                                 |
| evidence_screening_decision | Per-record include/exclude/uncertain decision.                                                        | Record link, reason, actor/model, timestamp, validation state.                         |
| evidence_extraction         | Candidate study characteristics and outcome/effect data.                                              | Exact source record, field provenance, extractor/model, reviewer status.               |
| evidence_appraisal          | Risk-of-bias or methodological appraisal inputs and candidate judgements.                             | Tool/framework version, model assistance, reviewer confirmation.                       |
| evidence_claim              | Atomic statement within a synthesis.                                                                  | Links to supporting, contradicting and non-informative evidence records.               |
| evidence_synthesis          | Versioned synthesis over a defined evidence set.                                                      | Evidence-set fingerprint, model/tool version, prompt/template version, reviewer state. |
| evidence_certainty_input    | Structured factors relevant to certainty assessment.                                                  | Explicit inputs and rationale; no automatic authoritative certainty label.             |
| evidence_translation        | Clinical or research-facing presentation layer.                                                       | Intended use, scope, date, limitations, human sign-off.                                |
| evidence_update_event       | New record, correction, retraction or material metadata change.                                       | Trigger source, affected synthesis version, review action and timestamp.               |

## 14.4 Review question and eligibility

The review question should be explicit before screening. PICO or related structured question models can be used where appropriate, but the schema should not force every diagnostic, prognosis or harms question into an intervention-only

template. Eligibility criteria should be stored and versioned so that later reviewers can reconstruct why records were included or excluded.

## 14.5 AI-assisted screening

AI may rank records, flag probable duplicates or propose screening decisions. Every excluded record that materially affects the evidence set should retain a reason and validation state. Screening automation should be validated prospectively on fixed evidence sets before it is trusted to hide low-ranked records from reviewers.

## 14.6 Structured extraction

AI may extract candidate study characteristics, outcomes, effect measures or methodological features. Critical values should remain traceable to the exact source record and, when full text is used lawfully, to a location or source fragment sufficient for human verification.

## 14.7 Methodological appraisal

AI can prepare structured inputs for risk-of-bias assessment or detect missing methodological information. It should not silently emit an authoritative risk-of-bias verdict. Appraisal frameworks are method-specific and often require judgement that remains accountable to a qualified reviewer.

## 14.8 Certainty and GRADE boundary

AI-EVIDENCE-02 may store structured inputs relevant to certainty assessment: risk of bias, inconsistency, indirectness, imprecision, publication bias and other applicable factors. It should not automatically declare an authoritative GRADE certainty rating or guideline recommendation without the methodology and human accountability required by GRADE.

## 14.9 Synthesis rules

- Distinguish direction of evidence from certainty of evidence.
- Represent contradictory studies and unresolved heterogeneity explicitly.
- Do not perform statistical pooling solely because numerical values can be combined.
- Record the exact evidence-set version/fingerprint used for each synthesis.
- Keep model, prompt/template and software version metadata for reproducibility.
- Preserve human edits as attributable revisions rather than overwriting the machine candidate.

## 14.10 Clinical translation

The final translation layer may present evidence relevant to a clinical question, with date, scope, limitations and provenance. It should not become a patient-specific prescription unless that intended use has been separately validated and governed. The user interface should allow the clinician to move from summary to claim and from claim to source evidence.

## 14.11 Living evidence updates

New trials, systematic reviews, corrections or retractions should create update events. When a material update changes the evidence base, a new synthesis version should be generated. Earlier versions should remain accessible so that past clinical or research decisions can be interpreted against the evidence available at that time.

## 14.12 Proposed validation states

| State           | Meaning                                                                              |
|-----------------|--------------------------------------------------------------------------------------|
| DRAFT_AI        | Machine-generated or machine-assisted candidate; not reviewed.                       |
| REVIEW_REQUIRED | Ready for qualified human review.                                                    |
| HUMAN_REVISED   | Reviewer changed the candidate; machine and human contributions remain attributable. |

| State                              | Meaning                                                                                        |
|------------------------------------|------------------------------------------------------------------------------------------------|
| VALIDATED_FOR_RESEARCH_USE         | Validated for the defined research workflow only.                                              |
| VALIDATED_FOR_DEFINED_CLINICAL_USE | Requires separate clinical safety, usability, governance and applicable regulatory validation. |
| REJECTED                           | Candidate rejected; reason preserved for audit.                                                |
| SUPERSEDED                         | Historical version retained but replaced by a newer validated/reviewed version.                |

### 14.13 PRISMA, Cochrane and GRADE relationship

AI-EVIDENCE-02 is methodologically inspired by systematic-review and evidence-certainty principles reflected in PRISMA, the Cochrane Handbook and GRADE. IML should use these as reference frameworks. An automated workflow is not automatically PRISMA-compliant, a Cochrane review or a GRADE assessment. Formal claims require the corresponding methods, reporting and human accountability.

## 15 • Security, Privacy, Licensing and Failure Containment

### 15.1 Secrets

Passwords, institutional cookies, API secrets and access tokens must not be stored in clinical evidence tables. Deployment-specific secret management should inject credentials at runtime with least privilege and rotation procedures.

### 15.2 Minimum necessary information

Evidence searches should transmit only de-identified, purpose-necessary information. Operational logs should not reintroduce the raw narrative that the evidence layer deliberately removed.

### 15.3 Access control

Clinical context, evidence queries and derived synthesis should be protected by role- and purpose-aware access controls. A researcher may be authorised to use a de-identified evidence set while lacking access to the linked patient encounter.

### 15.4 Rate limits and external dependency failure

Connectors should implement timeouts, retry/backoff, rate-limit handling and circuit-breaking as appropriate. Remote evidence-source failure should degrade the evidence workflow explicitly without blocking the core clinical record.

### 15.5 Licensing

Open metadata does not imply unrestricted rights to publisher full text. Every connector should preserve source-specific licence notes, open-access status where known and lawful link-out paths. Institutional access must be mediated without copying institutional credentials into the evidence database.

### 15.6 Retention and deletion

Retention should distinguish durable bibliographic provenance from transient raw responses, caches and model prompts. If raw provider responses are retained for debugging, their lifetime and privacy/licensing basis should be explicit.

### 15.7 AI prompt and output logging

For AI-EVIDENCE-02, reproducibility may require prompt/template version and hashes or controlled snapshots of model inputs/outputs. Raw prompts containing sensitive clinical material should not be logged indiscriminately. The provenance design should capture enough for review while minimizing exposure.

## 16 • Testing and Validation Roadmap

| Validation area            | Minimum reference tests before status advancement                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Database migrations        | Clean-install migration; sequential upgrade; re-run/idempotence behaviour; rollback/recovery plan; constraints; migration failure integrity.       |
| Backup/restore             | Backup generation, encrypted storage where required, restoration to clean environment, integrity checks and application restart.                   |
| Terminology                | Release import reproducibility, versioning, mapping ambiguity, provenance, translation review and expert validation.                               |
| HPRIM laboratory connector | Parsing, patient matching, units, reference ranges, source provenance, duplicate detection, correction, malformed-message handling.                |
| Other connectors           | Contract tests, timeout/partial-failure states, conformance, authentication, logging and source correction.                                        |
| Offline operation          | Network loss, local reads/writes, queueing, stale-data indication, conflict handling, resynchronisation and recovery.                              |
| Security                   | Threat model, least privilege, secret management, audit, patching, encryption where applicable, recovery and dependency vulnerability review.      |
| AI-EVIDENCE-01             | Source contract tests, de-identification, zero-result/failure distinction, deduplication, identifier integrity, retraction/update handling.        |
| AI-EVIDENCE-02 screening   | Fixed evidence-set benchmark, missed-study rate, false exclusion rate, disagreement analysis, reviewer workload and reproducibility.               |
| AI-EVIDENCE-02 extraction  | Field-level accuracy, source traceability, hallucination rate, reviewer correction burden and model-version sensitivity.                           |
| AI-EVIDENCE-02 appraisal   | Agreement with qualified human review, disagreement transparency, framework-specific validity and prohibition of unsupported authoritative labels. |
| Clinical translation       | Scenario-based safety review, uncertainty communication, source drill-down, human override and documented intended use.                            |
| Country packs              | Local terminology, identity, messaging, reimbursement/regulatory constraints and professional workflow validation.                                 |

## 16.1 Advancement rule

No component should advance in public status because a developer demonstration succeeded. Advancement requires pre-defined acceptance criteria, reproducible test evidence and explicit scope. A connector may be technically validated while the clinical workflow remains unvalidated.

## 16.2 AI prospective comparison

AI-EVIDENCE-02 should be tested prospectively against conventional review workflows on fixed evidence sets. Outcomes should include missed studies, extraction accuracy, reviewer time, disagreement, provenance completeness, reproducibility and the effect of model updates.

## 16.3 Safety case

For any defined clinical use, the technical evidence should be assembled into a safety case: intended use, hazards, mitigations, validation evidence, residual risks, user training, monitoring and incident/correction procedures. The level of formality should match applicable regulatory obligations and clinical consequence.

# 17 • Current Implementation Status Matrix

| Component                            | Status                              | Current technical interpretation                                                                                                                       |
|--------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open Clinical Workspace architecture | REFERENCE IMPLEMENTATION WORKSTREAM | Common core, integration layer, country packs, PostgreSQL foundation and AI service layer are defined and partially represented in code/database work. |
| PostgreSQL clinical foundation       | REFERENCE IMPLEMENTATION WORK       | Existing clinical/identity schemas are referenced by current migrations; ongoing model consolidation remains required.                                 |
| Terminology canonicalisation         | REFERENCE IMPLEMENTATION WORK       | Migration 191 defines conservative canonicalisation and legacy preservation.                                                                           |
| KSH97-P -> ICD-10 mapping work       | ACTIVE WORKSTREAM                   | Progressive mapping remains inspectable and review-based.                                                                                              |
| ICD-11 readiness                     | SCAFFOLD / FUTURE WORK              | No claim of live ICD-11 clinical coding.                                                                                                               |
| France country pack                  | ACTIVE WORKSTREAM                   | Scope defined; HPRIM is first controlled integration target.                                                                                           |
| Guatemala country pack               | ACTIVE WORKSTREAM                   | Spanish adaptation and constrained-connectivity deployment are first objectives.                                                                       |
| Local structured clinical extraction | ACTIVE/REFERENCE WORKSTREAM         | Candidate fact extraction for clinician review; no silent chart insertion.                                                                             |
| AI service layer                     | ARCHITECTURE + SCAFFOLD             | Models are assistive and replaceable; durable clinical record remains independent.                                                                     |

| Component                                           | Status                           | Current technical interpretation                                                                                             |
|-----------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| AI-EVIDENCE-01                                      | SCAFFOLD v0.1.1                  | PostgreSQL schema and source registry implemented; external connectors intentionally disabled pending validation.            |
| PubMed connector                                    | PLANNED/VALIDATION REQUIRED      | Source contract and API endpoint defined; operational validation required.                                                   |
| Europe PMC connector                                | PLANNED/VALIDATION REQUIRED      | Source contract and API endpoint defined; operational validation required.                                                   |
| ClinicalTrials.gov connector                        | PLANNED/VALIDATION REQUIRED      | Source contract and API endpoint defined; operational validation required.                                                   |
| Crossref connector                                  | PLANNED/VALIDATION REQUIRED      | Source contract and API endpoint defined; operational validation required.                                                   |
| Cochrane public metadata connector                  | PLANNED/VALIDATION REQUIRED      | Metadata/link-out only; full-text rights remain separate.                                                                    |
| AI-EVIDENCE-02                                      | PROPOSED                         | Appraisal/synthesis/translation object model defined in the current manuscripts; no implementation/clinical claim.           |
| Generative patient-specific clinical recommendation | OUTSIDE CURRENT VALIDATED STATUS | Would require separate intended-use definition, clinical safety, usability, governance and applicable regulatory validation. |

## 18 • Near-Term Technical Roadmap

1. Consolidate the PostgreSQL clinical model and document canonical schema contracts.
2. Complete and test terminology canonicalisation, KSH97-P encounter-reason workflows and progressive ICD-10 mappings.
3. Implement and validate the first France HPRIM laboratory-result pathway with source-preserving ingestion and correction.
4. Keep the Guatemala pack aligned with the common clinical model while testing Spanish language and constrained-connectivity deployment.
5. Implement application-side AI-EVIDENCE-01 connector clients one source at a time, beginning with reproducible contract tests and de-identification controls.
6. Add canonical deduplication, retraction/update checks and source-policy-aware metadata caching.
7. Define the AI-EVIDENCE-02 SQL/API contract separately from AI-EVIDENCE-01 rather than expanding retrieval tables into opaque synthesis blobs.
8. Build a fixed evidence-set validation harness for screening, extraction, claim lineage and reviewer disagreement.

9. Implement a clinician/researcher evidence UI that allows source drill-down and distinguishes machine candidate from human-validated synthesis.
10. Run security, offline/resilience, backup/restore and deployment tests across reference operating environments.
11. Advance module status only when pre-defined validation evidence exists.

## 18.1 Recommended AI-EVIDENCE-02 implementation sequence

1. Specification of objects, state machine and provenance fields.
2. Database migration in additive form with no changes required to AI-EVIDENCE-01 records.
3. Non-AI baseline workflow for manual screening/extraction/appraisal so that later AI assistance has a comparator.
4. AI-assisted candidate generation with full provenance and reviewer controls.
5. Fixed-set technical validation and error analysis.
6. Research-use validation in a defined evidence-review workflow.
7. Only after separate safety/regulatory assessment, consideration of a defined clinical-facing use.

## Appendix A • AI-EVIDENCE-01 Field-Level Summary

| Table                   | Important fields or constraints                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| evidence_source         | source_code; name; source_kind; access_mode; homepage_url; api_base_url; enabled; stores_full_text; capabilities JSON; licence_note; timestamps.                                       |
| evidence_query          | id; encounter_id; assessment_id; question_sha256; question_type; language_code; search_expression; search_expression_sha256; filters; status; requested_by; requested/completed times. |
| evidence_source_run     | query_id; source_code; run_status; request/response fingerprints; result_count; error_code/error_summary; started/completed times; metadata.                                           |
| evidence_record         | canonical_key; title; record_type; source_title; publication_date/year; abstract_available; open_access; retracted; metadata; timestamps.                                              |
| evidence_identifier     | record_id; source_code; identifier_type/value; source_url; retrieved_at; raw_metadata_sha256; uniqueness by source/type/value.                                                         |
| evidence_query_result   | query_id; source_run_id; record_id; source_rank; relevance_score; retrieval_status; exclusion_reason; retrieved_at; metadata.                                                          |
| evidence_access_profile | profile_code; institution_name; source_code; access_mode; enabled; configuration; timestamps; uniqueness by profile/source.                                                            |



## Appendix B • AI-EVIDENCE-02 Proposed Field Contract

| Object                      | Candidate fields for a future additive migration                                                                                                                          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| evidence_review_question    | id, query_id or linked question, framing_type, population/intervention/comparator/outcomes JSON, scope, intended_use, version, fingerprint, created_by, timestamps.       |
| evidence_eligibility_rule   | id, review_question_id, rule_type, expression/text, version, active flag, created_by, timestamps.                                                                         |
| evidence_screening_decision | review_question_id, record_id, decision, reason_code/text, actor_type, actor_id/model_run_id, validation_state, reviewer_id, timestamps.                                  |
| evidence_extraction         | review_question_id, record_id, extraction_schema_version, field_key, candidate_value JSON, source_locator, actor/model provenance, reviewer status.                       |
| evidence_appraisal          | review_question_id, record_id, framework_code/version, domain_key, candidate_judgement, rationale, reviewer judgement, provenance.                                        |
| evidence_claim              | synthesis_id, claim_text, claim_type, support_direction, confidence/uncertainty metadata, provenance.                                                                     |
| evidence_claim_record       | claim_id, record_id, relation_type<br>supporting/contradicting/non_informative, rationale, reviewer status.                                                               |
| evidence_synthesis          | review_question_id, evidence_set_fingerprint, synthesis_version, model/tool provenance, prompt/template version, narrative/structured output, validation_state, reviewer. |
| evidence_certainty_input    | synthesis/claim scope, factor code, candidate assessment, rationale, reviewer assessment; no automatic final GRADE claim.                                                 |
| evidence_translation        | synthesis_id, audience, intended_use, summary, limitations, validation state, sign-off actor/date.                                                                        |
| evidence_update_event       | source event, affected record/synthesis, event type, detected_at, action state, resulting synthesis version.                                                              |

*The field contract is deliberately proposed rather than normative. The exact schema should be reviewed against implementation experience, systematic-review methodology and privacy/security requirements before a migration is committed.*

## Appendix C • Evidence Source Registry and Reference Endpoints

| Code               | Source                           | Role                                    | Reference endpoint / access pattern                                                                               |
|--------------------|----------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| PUBMED             | PubMed / NCBI                    | Biomedical bibliographic discovery      | <a href="https://eutils.ncbi.nlm.nih.gov/entrez/eutils/">https://eutils.ncbi.nlm.nih.gov/entrez/eutils/</a>       |
| EUROPE_PMC         | Europe PMC                       | Bibliographic metadata and OA discovery | <a href="https://www.ebi.ac.uk/europepmc/webservices/rest/">https://www.ebi.ac.uk/europepmc/webservices/rest/</a> |
| CLINICALTRIALS_GOV | ClinicalTrials.gov               | Trial registry/results-status metadata  | <a href="https://clinicaltrials.gov/api/v2/">https://clinicaltrials.gov/api/v2/</a>                               |
| CROSSREF           | Crossref                         | DOI-centred scholarly metadata          | <a href="https://api.crossref.org/">https://api.crossref.org/</a>                                                 |
| COCHRANE_PUBLIC    | Cochrane Library public metadata | Systematic-review discovery/link-out    | Public web/metadata access; subscription/full-text rights handled separately.                                     |

## Appendix D • Primary Documentation and Methodological References

- NCBI Entrez Programming Utilities: <https://www.ncbi.nlm.nih.gov/books/NBK25501/>
- Europe PMC RESTful Web Service: <https://europepmc.org/RestfulWebService>
- ClinicalTrials.gov API documentation: <https://clinicaltrials.gov/data-api/api>
- Crossref REST API documentation: <https://www.crossref.org/documentation/retrieve-metadata/rest-api/>
- Cochrane Library: <https://www.cochrane.org/products-and-services/cochrane-library>
- Cochrane Handbook for Systematic Reviews of Interventions: current online edition.
- PRISMA 2020 statement and associated checklist/explanation resources.
- GRADE Working Group guidance and GRADE Book: current online resources.
- WHO. Ethics and governance of artificial intelligence for health.
- NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0).
- HL7 FHIR documentation, openEHR specifications, DICOM standard and LOINC documentation as applicable to implementation modules.
- Harrison's Principles of Internal Medicine. Current licensed edition available to the participating institution or clinician; chapter and edition should be recorded when used.
- Goldman-Cecil Medicine. Current licensed edition available to the participating institution or clinician; chapter and edition should be recorded when used.
- Current Medical Diagnosis & Treatment (CMDT). Current licensed edition; chapter/section and edition should be recorded when used.
- IML clinical evidence methodology: clinical question -> current guidelines -> Harrison/Cecil/CMDT licensed clinical synthesis -> systematic reviews/meta-analyses -> recent primary literature -> comparative synthesis -> clinician review.